

Dyne Lee

Cambridge, MA | dyne@mit.edu | (408)-202-3893
Website | LinkedIn

EDUCATION

- **MIT** (Sept. 2023 - May 2027): B.S. in Mathematics and Computer Science | GPA 5.0
 - Current Coursework: Number Theory, Cryptography

EXPERIENCE

- **Geometry Research at Imperial College London** June 2026 - August 2026
Undergraduate Researcher
 - Studied spin structures on homogeneous spaces, supervised by Dr. Marie-Amélie Lawn.
 - Classified quaternionic spin structures on quaternion-Kähler symmetric spaces, using techniques from representation theory and Lie theory.
- **MIT Summer Program in Undergraduate Research** June 2025 - Present
Undergraduate Researcher
 - Studied conormal bundles to certain orbits on Grassmannians and flag varieties, supervised by Katia Bogdanova and Prof. Roman Bezrukavnikov. Proved sufficient conditions for the closure of a conormal bundle to be a normal variety, using a resolution of singularities.
 - Constructed an analog of the Zelevinsky resolution of singularities for the closure of a conormal bundle.
 - Paper in progress. [Link to preliminary writeup](#)
- **MIT Directed Reading Program** Jan. 2025, Jan. 2026
Student
 - Completed a reading project on regular rings and applications to the proofs of the Auslander-Buchsbaum Theorem and the Chevalley-Shepard-Todd Theorem.
 - Completed a reading project on representations of $SL_2(F_q)$ using l -adic cohomology of the Drinfeld curve.
- **Algebraic Geometry Research at MIT** Dec. 2024 - May 2025
Undergraduate Researcher
 - Conducted research on moduli spaces of quiver representations through the MIT Undergraduate Research Opportunities Program, supervised by Dr. Miguel Moreira.
 - Studied descendent algebras associated to quivers and functionals on these algebras satisfying constraints from the action of the Virasoro algebra. Applied concepts from algebraic combinatorics to connect this question to the language of symmetric functions.
- **Quantitative Research at MIT Sloan School of Management** June 2024 - Present
Undergraduate Researcher
 - Implemented and trained Alternating Least Squares and stochastic gradient descent algorithms for matrix completion, along with a Bayesian prior, on CRSP market data.
 - Extended algorithms to time series data with sliding and expanding window algorithms and autoregressive moving average models.
 - Applied results to improve a covariance structure factor model for assets.

PROJECTS

- **MIT Pokerbots** Dec. 2024 - Feb. 2024
Team Lead
 - Collaborated with teammates to build a computer bot to compete against others in a variant of poker.
 - Used machine learning, Counterfactual Regret Minimization (CFR), k-means clustering, and sampling techniques to compute abstractions of the game tree and calculate equilibrium strategies.
 - Awarded 1st place in Java, 9th place overall

COMPLETED COURSEWORK

- **Mathematics:** Algebraic Geometry, Lie Theory, Representation Theory, Functional Analysis, Geometry of Manifolds, Combinatorial Theory, Algebraic Groups, Noncommutative Algebra
- **Computer Science:** Theory of Computation, Machine Learning, Design and Analysis of Algorithms

SKILLS

- **Mathematics:** Linear algebra, statistics, probability, statistical learning, analysis
- **Computer Science:** Python (NumPy, scikit, pandas), Java, PyTorch, TensorFlow SQL, Mathematica
- **Languages:** English (native), Korean (native), Spanish (proficient), Mandarin (proficient)